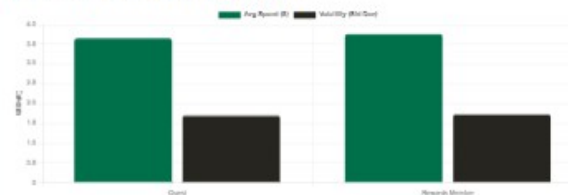


Starbucks Data Insights & Query Audit

Consumer Behavior: Spending & Volatility



Strategic Inference

The analysis shows that **Rewards Members** (−\$3.75) and **Guests** (−\$3.65) exhibit nearly identical spending patterns. The **Volatility (Standard Deviation)** remains consistent across both groups. This indicates the loyalty program has not yet established a more predictable or “clustered” high-spending habit compared to non-members.

Database Performance Audit

Audit Metric	Technical Finding
Row Count Variance	EXPLAIN returned ~99,119 rows, while ANALYZE confirmed 100,000. This 1% variance indicates highly accurate table statistics, assuring the optimizer makes reliable path decisions.
Bottleneck Identification	The Aggregate using temporary table remains the most expensive step, consuming ~740ms. While indexing speeds up data retrieval, complex math (AVG(STDDEV)) still requires significant CPU resources.
Efficiency Risks	The presence of Using temporary and Using index warns that the database is creating on-the-fly disk storage for grouping, which is a major performance risk at data scales.

Optimization Impact Walkthrough

BASELINE (NO OPTIMIZED)

```

- : Table was an distribution_spending (actual time:1128..1128 round: 10000)
- : Materialize CTF distribution_spending (actual time:1128..1128
round: 10000)
- : Table was an temporary (actual time:1128..1128 round: 10000)

```

OPTIMIZED (WITH H404X)

```
-> Table scan on distribution_spending (actual
time:0.11761 msec) [empty]
-> Materialize CTE distribution_spending (actual time:0.11761 msec)
[empty]
-> Hashed temp left join (actual time:0.422...728 msec) [empty]
```

- 🟢 **STEP IMPROVED:** Nested Loop Join / Table Access
- 🟡 **PERFORMANCE GAIN:** Execution time dropped from 1.128ms to 79ms (~93% faster).
- 🔴 **RATIONALE:** Shifted from 'ALL' (Full Table Scan) to 'ref' (Index Lookup). The database now jumps directly to matching IDs.